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INDUSTRY

| COLLEGE OF ELECTRICAL AND MECHANICAL ENGINEERING.

| DEPARTMENT OF SOFTWARE ENGINEERING.

| DATABASE SYSTEM (SWEG2108).

PROJECT REPORT

TITLE: HOSPITAL MANAGEMENT SYSTEM(HMS).

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Declaration

We hereby declare that this project entitled "Hospital Management System (HMS): Hybrid MySQL & MongoDB Healthcare Database Solution" is the original work of our group and has not been submitted to any other institution or organization for academic evaluation.

All references and sources used in this report have been properly acknowledged.

Acknowledgment

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Abstract

Healthcare institutions generate and process large amounts of critical information daily, including patient records, appointment schedules, medical histories, billing transactions, emergency cases, and departmental operations. Traditional manual systems and poorly organized digital systems often result in inefficiency, data redundancy, delayed services, and poor information management.

This project presents the design and implementation of a Hospital Management System (HMS) using both relational and NoSQL database technologies. The system integrates MySQL for structured relational data management and MongoDB for flexible document-oriented data storage.

The project includes problem analysis, entity relationship modeling, logical schema design, normalization up to Boyce-Codd Normal Form (BCNF), and practical implementation using SQL and MongoDB queries. The system supports major healthcare operations such as patient registration, appointment scheduling, billing management, emergency case handling, queue management, and medical record processing.

The implementation demonstrates how hybrid database systems can improve healthcare data organization, consistency, scalability, and operational efficiency.

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Organization

Hospitals are among the most data-intensive organizations in modern society. Every day, healthcare institutions manage large amounts of information related to patients, doctors, appointments, treatments, prescriptions, billing activities, emergency cases, and medical histories. Accurate and organized management of this information is essential for effective healthcare delivery.

Many healthcare institutions, particularly in developing environments, still rely on manual systems or partially digitized solutions for handling medical records and administrative operations. Such systems often result in inefficiency, delayed services, duplication of records, poor coordination among departments, and difficulties in retrieving important patient information.

The advancement of database technologies has created opportunities for hospitals to improve operational efficiency and healthcare quality through computerized information systems. Modern Hospital Management Systems (HMS) enable centralized data storage, faster information retrieval, improved patient tracking, better appointment scheduling, efficient billing management, and enhanced decision-making.

This project focuses on the design and implementation of a Hospital Management System using both MySQL and MongoDB database technologies. The system aims to improve healthcare data management by integrating relational and NoSQL database approaches into a hybrid architecture.

1.2 Problem Statement

Many hospitals continue to face serious challenges in managing healthcare information efficiently. Traditional manual systems and poorly designed digital systems often lead to:

- Loss or duplication of patient records
- Long patient waiting times
- Poor appointment scheduling
- Inefficient queue management
- Delayed emergency handling
- Billing inconsistencies
- Difficulty accessing medical information
- Weak coordination between departments
- Data redundancy and inconsistency

These problems negatively affect operational efficiency, increase staff workload, reduce service quality, and may even endanger patient safety during critical situations.

Furthermore, some healthcare systems rely only on rigid relational databases that may struggle to efficiently handle flexible or semi-structured medical information. This creates limitations in scalability and adaptability.

Therefore, there is a need for a modern Hospital Management System capable of efficiently managing healthcare operations through a hybrid database architecture using both relational and NoSQL database technologies.

1.3 Objectives

1.3.1 General Objective

To design and implement a Hospital Management System using MySQL and MongoDB for efficient healthcare data management.

1.3.2 Specific Objectives

The specific objectives of the project are:

- To analyze healthcare data management challenges in hospitals
- To identify major entities and relationships within the hospital environment
- To design an Entity Relationship Diagram (ERD)
- To develop a logical database schema
- To normalize the database up to Boyce-Codd Normal Form (BCNF)
- To implement the relational database using MySQL
- To implement the NoSQL database using MongoDB
- To manage patient records, appointments, billing, prescriptions, and emergency cases efficiently
- To test and validate database functionality
- To improve healthcare information organization and accessibility

1.4 Scope of the Project

This project focuses on the design and implementation of a Hospital Management System database for managing core healthcare operations.

The system includes:

- Patient management
- Doctor management
- Appointment scheduling
- Medical record handling
- Billing and payment management

- Queue management
- Emergency case handling
- Department management
- Prescription management

The project covers database analysis, design, implementation, and testing using MySQL and MongoDB.

However, the project does not include:

- Full web application development
- Mobile application integration
- Real-time cloud deployment
- Advanced cybersecurity implementation
- Biometric authentication systems
- Machine learning integration

1.5 Benefits of the Project

The proposed Hospital Management System provides several benefits, including:

- Improved patient information management
- Faster appointment scheduling and retrieval
- Better emergency case prioritization
- Reduced data redundancy
- Enhanced healthcare service efficiency
- Organized billing and payment tracking
- Improved departmental coordination
- Better scalability through hybrid database integration
- Improved accuracy and consistency of medical data

1.6 Methodology and Tools

The project followed a structured database development methodology consisting of analysis, design, implementation, and testing phases.

Methodology

The following approaches were used during project development:

1. Requirement Analysis
2. Entity Identification
3. Relationship Modeling
4. ER Diagram Design
5. Normalization

6. Database Implementation
7. Testing and Validation
8. Documentation

Tools Used

The following tools and technologies were used:

Tool	Purpose
MySQL Workbench	Relational database implementation
MongoDB Compass	NoSQL database implementation
Draw.io / Lucidchart	ER diagram design
GitHub	Version control and collaboration
Canva / PowerPoint	Presentation preparation
Google Docs	Documentation

CHAPTER TWO

DATABASE DESIGN

2.1 Introduction

Database design is one of the most important stages in developing an efficient information management system. A properly designed database ensures data consistency, minimizes redundancy, improves retrieval efficiency, and supports scalability.

This chapter presents the design of the Hospital Management System database, including entity identification, relationship modeling, ER diagram development, logical schema creation, and normalization up to Boyce-Codd Normal Form (BCNF).

2.2 Entity Identification

The following entities were identified for the Hospital Management System:

Entity	Description
Patient	Stores patient information
Doctor	Stores doctor information
Department	Stores hospital department data
Appointment	Manages appointment scheduling
Medical_Record	Stores patient diagnosis and treatment
Prescription	Stores prescribed medicines
Billing	Stores billing information
Payment	Stores payment transactions
Queue_System	Manages patient waiting queues
Emergency_Case	Handles emergency patient cases

2.3 Attribute Design

Patient

- patient_id (PK)
- full_name
- gender
- date_of_birth
- phone
- address
- blood_group

Doctor

- doctor_id (PK)
- full_name
- specialization
- phone
- department_id (FK)

Department

- department_id (PK)
- department_name
- location

Appointment

- appointment_id (PK)
- patient_id (FK)
- doctor_id (FK)
- appointment_date
- appointment_time
- status

Medical_Record

- record_id (PK)
- patient_id (FK)
- diagnosis
- treatment
- record_date

Prescription

- prescription_id (PK)
- record_id (FK)
- medicine_name
- dosage
- duration

Billing

- bill_id (PK)
- patient_id (FK)
- total_amount
- bill_date

Payment

- payment_id (PK)
- bill_id (FK)
- payment_method
- payment_date
- amount_paid

Queue_System

- queue_id (PK)
- appointment_id (FK)
- queue_number
- priority_level
- estimated_time

Emergency_Case

- emergency_id (PK)
- patient_id (FK)
- severity_level
- arrival_time

2.4 Relationship Design

The relationships between entities are as follows:

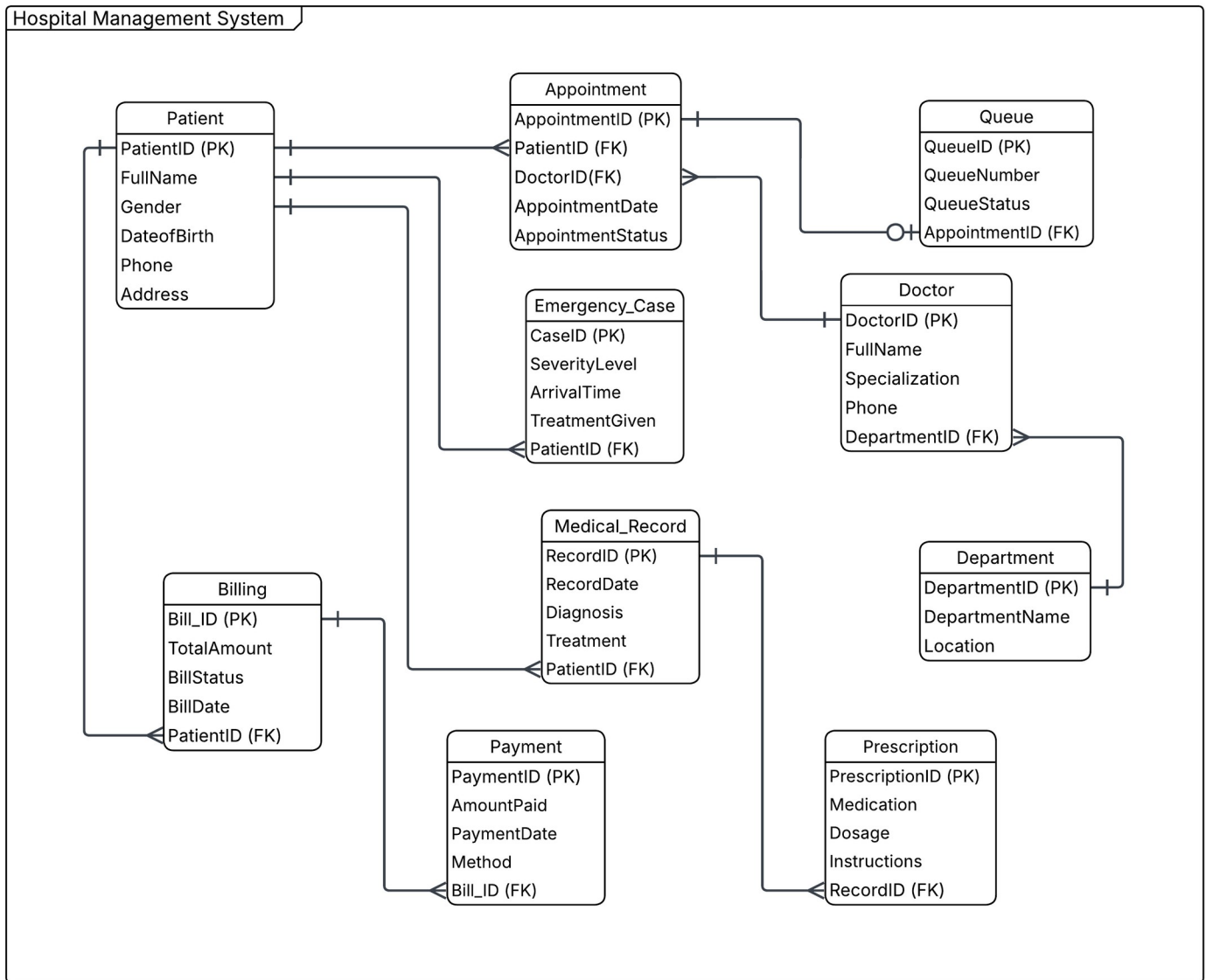
Relationship	Type
Patient books Appointment	1:M
Doctor handles Appointment	1:M
Doctor belongs to Department	M:1
Patient has Medical_Record	1:M
Medical_Record has Prescription	1:M
Patient generates Billing	1:M
Billing has Payment	1:M
Appointment has Queue_System	1:1
Patient may have Emergency_Case	1:M

2.5 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) visually represents the entities, attributes, and relationships within the Hospital Management System.

The ERD includes:

- Primary keys
- Foreign keys
- One-to-many relationships
- One-to-one relationships
- Relational constraints



2.6 Logical Schema Design

The ER Diagram was transformed into relational tables with clearly defined primary keys and foreign keys.

The logical schema supports:

- Referential integrity
- Efficient data retrieval
- Relationship consistency
- Scalable database management

2.7 Database Normalization

The database schema was normalized to eliminate redundancy and improve data integrity.

First Normal Form (1NF)

The database tables were structured to ensure:

- Atomic attributes
- Elimination of repeating groups
- Single-valued fields

Second Normal Form (2NF)

Partial dependencies were removed by ensuring that all non-key attributes depend entirely on the primary key.

Third Normal Form (3NF)

Transitive dependencies were removed to improve consistency and reduce redundancy.

Boyce-Codd Normal Form (BCNF)

The database was further normalized to BCNF to ensure that every determinant is a candidate key.

Benefits achieved through normalization include:

- Reduced redundancy
- Improved consistency
- Better dependency management
- Efficient storage organization
- Easier maintenance

CHAPTER THREE

IMPLEMENTATION AND TESTING

3.1 Introduction

This chapter presents the implementation and testing of the Hospital Management System database using MySQL and MongoDB technologies.

The implementation phase transformed the conceptual and logical designs into functional database systems capable of handling healthcare information efficiently.

3.2 MySQL Implementation

3.2.1 Database Creation

The relational database was implemented using MySQL Workbench.

Example:

```
CREATE DATABASE HMS_DB; USE HMS_DB;
```

3.2.2 Table Creation

The following tables were created:

- Patient
- Doctor
- Department
- Appointment
- Medical_Record
- Prescription
- Billing
- Payment
- Queue_System
- Emergency_Case

Primary keys and foreign keys were implemented to maintain relational integrity.

3.2.3 Sample SQL Query

```
SELECT Patient.full_name, Doctor.full_name, Appointment.appointment_date FROM Appointment JOIN  
Patient ON Appointment.patient_id = Patient.patient_id JOIN Doctor ON Appointment.doctor_id =  
Doctor.doctor_id;
```

3.3 MongoDB Implementation

MongoDB was implemented as the NoSQL component of the system to support flexible healthcare data storage.

Collections Created

- patients
- doctors
- appointments
- medical_records
- billing
- emergency_cases

Sample MongoDB Document

```
{ "patient_id": 1, "full_name": "Abel Tesfaye", "gender": "Male", "blood_group": "O+", "phone": "0911223344" }
```

Sample MongoDB Query

```
db.patients.find()
```

3.4 Testing and Validation

System testing was conducted to verify:

- Correct table creation
- Relationship functionality
- Query execution
- Constraint enforcement
- Data consistency
- Record retrieval accuracy

Testing was performed using sample hospital data scenarios.

3.5 Results

The implementation demonstrated successful:

- Patient information management
- Appointment scheduling
- Medical record handling
- Billing and payment management

- Queue handling
- Emergency case prioritization
- Relational and NoSQL integration

The system achieved improved organization, reduced redundancy, and efficient healthcare information management.

3.6 Challenges Encountered

The following challenges were encountered during implementation:

- Designing complex entity relationships
- Maintaining BCNF normalization
- Managing foreign key dependencies
- Understanding MongoDB document structures
- Query optimization
- Integrating relational and NoSQL systems

3.7 Future Improvements

Potential future improvements include:

- AI-assisted healthcare analysis
- Mobile application integration
- Cloud deployment
- Real-time appointment notifications
- Advanced cybersecurity mechanisms
- Real-time analytics dashboards

Conclusion

The Hospital Management System successfully demonstrated the practical application of relational and NoSQL database technologies in solving healthcare data management challenges.

The integration of MySQL and MongoDB enabled efficient handling of patient records, appointments, billing, queue systems, medical records, and emergency case management.

The project also demonstrated the importance of proper database design, normalization, and hybrid database architecture in developing scalable and reliable healthcare systems.

Overall, the system provides a strong foundation for future healthcare information management solutions.

Recommendations

Based on the project findings, the following recommendations are proposed:

- Hospitals should adopt digital healthcare management systems to improve operational efficiency.
- Healthcare institutions should integrate both relational and NoSQL databases for flexible and scalable information management.
- Future systems should include cloud deployment and mobile accessibility.
- Advanced security and data encryption mechanisms should be integrated into healthcare databases.
- AI-assisted analytics can further improve decision-making and patient prioritization.

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ANNEXES

Annex A — Sample Survey Questions

1. How are patient records currently managed?
2. What challenges exist in appointment scheduling?
3. How are billing and payment records stored?
4. What problems occur during emergency case handling?
5. Is the current system fully computerized?
6. What improvements are needed in healthcare information management?

Annex B — Sample Forms

Patient Registration Form

- Patient ID
- Full Name
- Gender
- Date of Birth
- Phone Number
- Address
- Blood Group

Appointment Form

- Appointment ID
- Patient ID
- Doctor ID
- Appointment Date
- Appointment Time
- Status

Annex C — Sample SQL Code

```
CREATE TABLE Patient ( patient_id INT PRIMARY KEY AUTO_INCREMENT, full_name VARCHAR(100),  
gender VARCHAR(10), date_of_birth DATE, phone VARCHAR(20), address VARCHAR(200), blood_group  
VARCHAR(5) );
```

Annex D — GitHub Repository Structure

/docs final_report.pdf presentation.pptx /mysql schema.sql queries.sql /mongodb collections.json
queries.js /diagrams erd.png normalization.pdf /annex survey_questions.pdf sample_forms.pdf
CONTRIBUTION.md README.md